



The student's behavior and attitudes toward using AI applications in their education

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1. Abstract

Our aim is to understand how students behave and feel about using AI applications in their education. This study looks at the reasons behind their acceptance or hesitation, how they interact with AI tools, and what they expect from them. By studying their experiences and challenges, we hope to find useful insights that can help improve the use of AI in learning. The results will support creating better AI-based tools that make education more effective and .personalized

2.Introduction

in our research we will focus on targeting university students and asking them about "using artificial intelligence applications during their education" Artificial intelligence applications: It is an artificial intelligence bot that allows you to have human-like conversations such as answering common questions, writing articles, scientific research, emails, advertising and more. Student Attitudes: Perceptions of AI as a helpful or disruptive technology. Concerns related to AI in education, such as data privacy, ethical concerns, and trustworthiness. Positive attitudes, such as excitement about innovation personalization. AI in Education: Personalization of learning experiences. Improved accessibility and inclusivity for students with diverse needs. Time-saving for both students and educators. In our research, we will make a comparison between students who use artificial intelligence applications during their studies and other students who do not use them, so we will come up with a number of questions such as: Do AI applications make a difference during the learning process? The importance of our research lies in the study of the student's method when he uses artificial intelligence applications during education and whether there is a difference when the student uses artificial intelligence programs during his years of education or not.

3.Problem statement

Our aim is to understand the students' behaviour and attitudes toward using AI applications in their education.

4.Objectives

Our goal in this research is to see the technological development that led to the development of artificial intelligence technologies and applications in several fields and time intervals before and after the use of artificial intelligence, and we will seek to solve the problems of complete reliability in some students when they use these applications for the purpose of homework or test solutions. Is the student's productivity affected when he uses artificial intelligence applications when doing homework or research, for:

1- Enhance Personalized Learning Utilize AI algorithms to analyses learner data and tailor educational content to individual needs, optimising learning outcomes and engagement.

2- Develop Intelligent Tutoring Systems Implement AI-powered systems to provide real-time, personalized guidance

and support, simulating one-on-one tutoring to address learners' specific challenges.

3- Leverage Natural Language Processing (NLP) Use NLP tools for automated language assessments, instant feedback, and chatbots to enhance learner communication, support, and writing skills development.

4- Enable Intelligent Content Creation Automate the creation of high-quality interactive learning materials through AI, saving time for educators and providing learners with diverse, up-to-date resources.

5- Apply Learning Analytics Analyze learner data to track progress, identify areas for improvement, and make data-driven decisions to enhance instructional strategies and learning outcomes.

6- Streamline Assessment and Feedback Use AI-powered tools to automate evaluation, provide immediate and personalised feedback, and implement adaptive testing for optimized learner engagement.

7- Utilize Predictive Analytics for Early Intervention Identify at-risk learners through predictive analytics and provide targeted support to improve engagement, retention, and overall success.

5.Recruitment brief

Date of research	• 8 December 2024
Location	• Online
Length of the research session	• 5-7 minutes
The number of the participants needed	• 50-70
The times at which you want to schedule	• 11 am – 3 pm
A list demographics and characteristics you want the participants to have	• Age: Students of educational age (for example, 18-22 years). • Gender: Both (male and female) • Educational level: university students • University major: computer • Location: local

6. Recruitement screener

Introduction	• This research aims to explore university students' behaviors and attitudes toward using AI applications in education, such as ChatGPT and other interactive learning tools. • The goal is to understand how students interact with these technologies and their impact on their educational experience.
Logistics and timing	• Date: December 8, 2024. • Location: Online. • Session Length: 5-7 minutes. • Number of Participants: 50-70. • Timing: Between 11:00 AM and 3:00 PM.
Demographics	• Age Group: University students aged 18-22 years. • Gender: Both males and females. • Educational Level: Undergraduate students. • Major: Computer-related fields (e.g., Computer Science or IT). • Location: Local students
Behaviour	• Some students regularly use AI tools for studying and completing assignments. • Others may hesitate to use these tools due to unfamiliarity or skepticism.
Confidence levels	• Evaluating students' trust in the accuracy and reliability of AI-generated responses. • Understanding whether they feel comfortable relying on these tools without further verification.
Technology and devices usage	• Availability of internet access and its impact on their ability to use AI applications. • Devices commonly used by students, such as smartphones and laptops.
Confidentiality and disclosures	• Questions about the privacy of the information they share with AI applications.
Accessibility	• Some students are concerned about the use of their data, while others do not care. • AI applications are generally accessible, but some students face challenges such as poor internet connectivity or lack of appropriate devices. • The availability of both free and paid AI tools provides diverse options for students.

7.Consent form

The purpose of this research project is to study the students behavior and attitudes toward using AI applications in their education. This is a research project being conducted by HCI students at UQU. You are invited to participate in this research project because you are a student in computer department . Your participation in this survey is voluntary. You may choose not to participate. If you do decide to participate, you may withdraw at any time. This online survey will take approximately (10) minutes to complete. Your responses will be confidential. We do collect identifying information such as (your name, your major). The survey questions will be about using AI applications in education . All data is stored in a password-protected electronic format. If you have any questions about this survey, please contact “hajiralharthi@gmail.com”.

please select your choice below: Clicking on the 'agree' button indicates that:

- You have read and understood the above information.
- You voluntarily agree to participate. If you do not wish to participate in the survey, please decline participation by clicking on the 'disagree' button.

<https://forms.office.com/r/EbEPqBv1kC>

8. The task

1. Executive Summary

Remote usability tests were conducted for the purpose of finding possible improvements in the product selection process.

The majority of participants understood how to use the product.

2. Methodology

Type of test: Moderated / Remote.

Screened participants using a general message

Connected with participants through “Teams meeting “

3.Participant Profiles

Our group of five testers consisted of young adults aged between 19-23 years old. All participants were female, and they used a variety of devices, including smartphones, laptops, and iPads, to engage with the test. Professionally, our testers represented a diverse range of experiences, aligning with the target demographic of the organization. Specifically, our testers identified as follows:

1-Reema, 19 years old, Smart phone

2-Ghadeer, 23 years old, Laptop

3-Raghad, 19 years old, Laptop

4-Shatha, 20 years old, Smart phone

5-Hala, 23 Years old, iPad

4.Task tested - Post Test questions

Tasks participants performed : Summarizing an article.

Data collection methods: Screen sharing, think-aloud.

How would you describe ChatGPT to someone?

What was your favorite aspect of ChatGPT ?

What was the most confusing part of the test?

Would you recommend this product to a friend or colleague?

5. Task Performance

We invited each participant separately in the Teams meeting, and asked them to share a screen so that we could monitor the results, and then made them open an application or ChatGPT website, after that we sent the article to them through a call conversation in Teams and asked them to send it to ChatGPT and ask it to summarize the article then we asked them the post-test questions. Results

6.Bugs and issues

- 1-Two participants faced the problem that the application summarized the article very briefly, contrary to the result they expected.
- 2-Ghadeer had a problem hanging the laptop while she was sharing the screen.

7.key findings and Recommendations

- Finding 1: Limited knowledge in ChatGPT.
Recommendation: Expanding knowledge of human demands.
- Finding 2: The lack of comprehensiveness of the audio recording service.
Recommendation: Development of the voice recording feature.
- Finding 3: limited features access.
Recommendation: Remove the limitation feature of questions.

8. Post-Test Feedback

- They all rate the ChatGPT 10 out of 10.
- The ability to communicate information.
- All participants were generally satisfied with the ChatGPT service.
- All of them described feeling comfortable, easy and positive while using ChatGPT.

9. Next Steps

- We suggest to the developers of ChatGPT to develop easy subscription plans for users.
- Develop the program so that it is able to understand human language and behavior.
- Collaborate with AI tool developers to improve user experience and ensure multilingual

9. Data analysis

Student questionnaire data on the use of the ChatGPT app was analyzed using Microsoft Forms. The analysis included calculating the percentages of answers and identifying repeated ideas and the participants' answers. A statistical analysis was also carried out to calculate the averages. After that, the data was organized into clear and concise reports to present the results in an orderly manner.

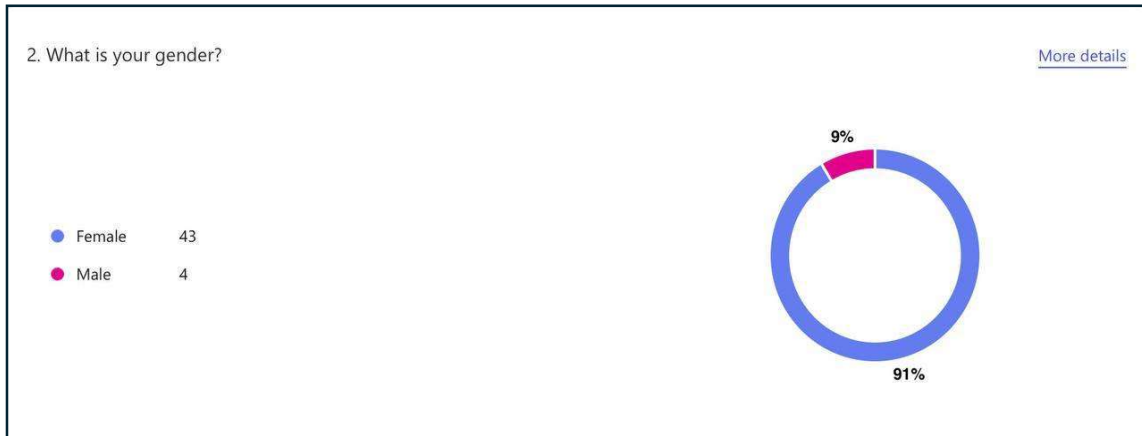
The analysis relied on quantitative and qualitative methods, where the data were analyzed digitally to determine ratios and statistics. Based on the results, some aspects that need improvement in the application have been identified, such as the use experience and the extent to which students' needs are met.

Figure1



In figure.1 90% of participants agreed to take part in the survey, indicating a high willingness to participate .Only 10% declined participation, showing minimal resistance or hesitation.

Figure 2



In figure.2 The majority of respondents (91%) are female, indicating a significant gender imbalance in the sample. Only 9% of participants are male, suggesting lower representation in this survey.

Figure 3



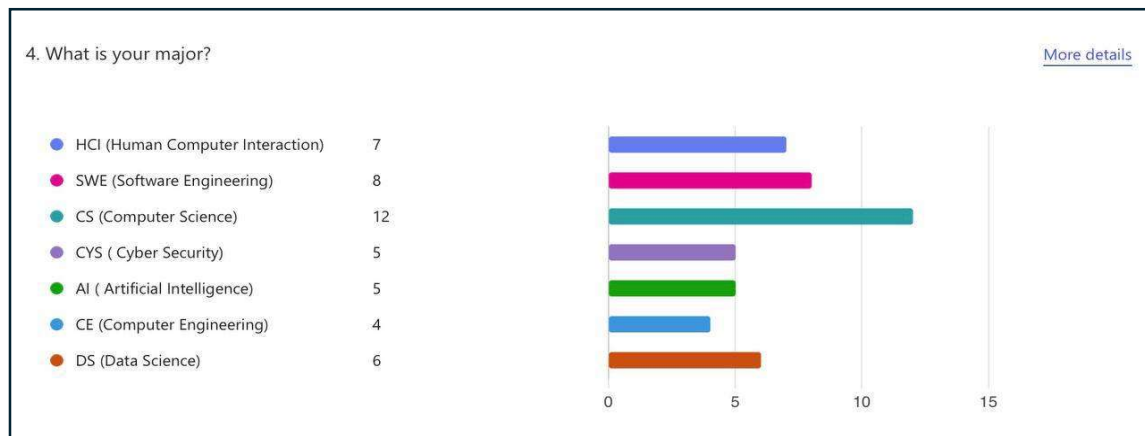
In figure.3 The majority of respondents (94%) are aged 19-23, indicating that most participants are likely university students.

Only 4% fall into the 24-28 age group, suggesting a small number of older students or early-career professionals.

No respondents are aged 29-33, reinforcing that the survey primarily targets younger individuals.

A single participant is under 18, implying minimal representation from high school students.

Figure4



In figure4

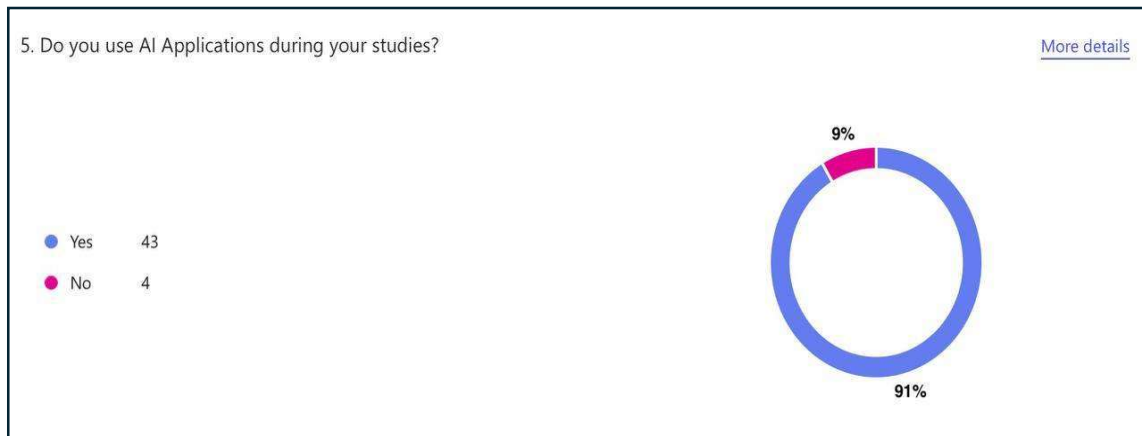
-Computer Science is the most common major (12 students), showing strong interest in core computing disciplines.

-Software Engineering (8) and Human-Computer Interaction (7) are also well-represented, indicating a focus on building and optimizing user-centered systems.

-Data Science (6) and Cyber Security (5) are equally significant fields, reflecting the growing importance of data-driven decision-making and security concerns.

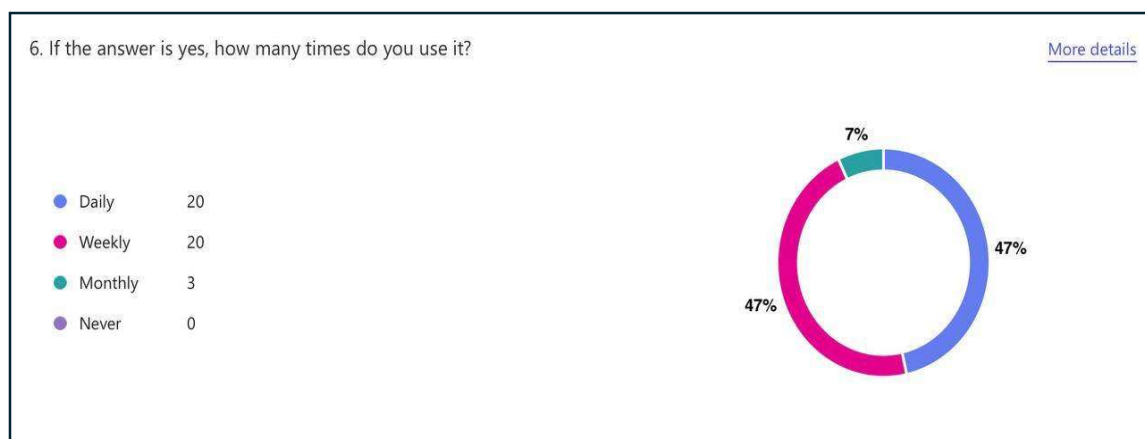
-Artificial Intelligence (5) and Computer Engineering (4) are slightly less common, but still have a presence among students.

Figure5



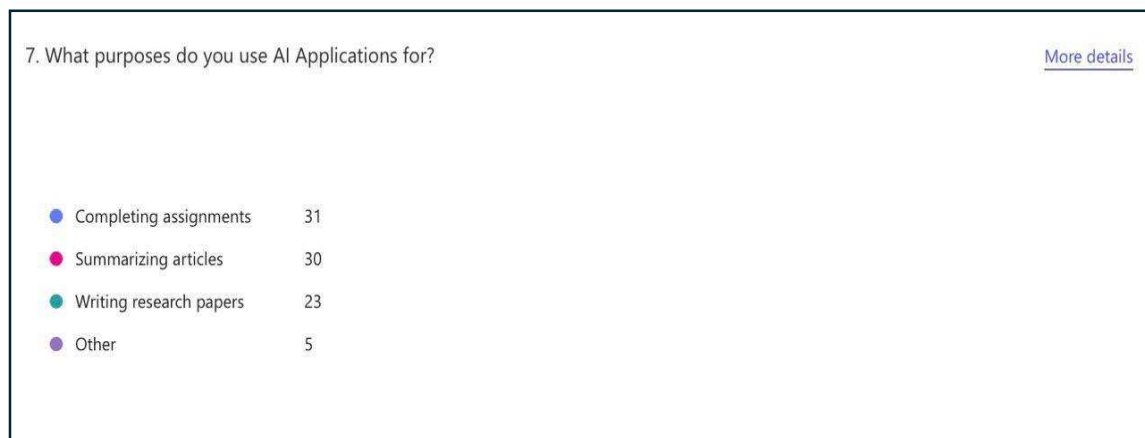
In figure.5 AI is widely used as a study aid, as evidenced by the fact that the vast majority of respondents (91%) use AI applications in their studies. It appears that non-users are a tiny minority, since only 9% do not use Ai program.

Figure6



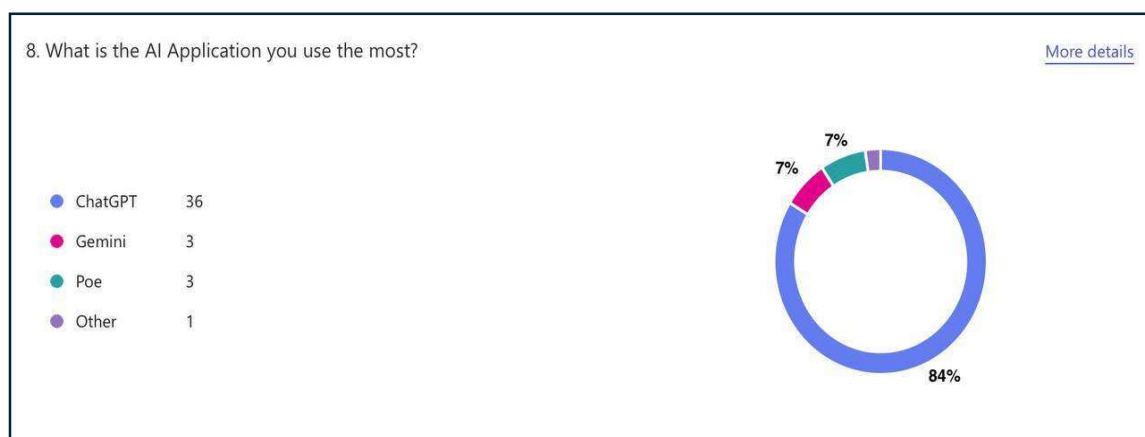
In figure.6 The vast majority (94%) utilize AI apps on a daily or weekly basis, suggesting that they are used often. Only 7% of people use AI apps on a monthly basis, indicating that infrequent use is uncommon. Since none of the respondents chose "Never" all of them utilize AI software occasionally.

Figure7



In figure.7 The most frequent applications include finishing assignments (31) and summarizing publications (30), suggesting that AI is mostly utilized to help with academic duties. A significant dependence on AI for academic writing is indicated by the fact that 23 users use it to write research papers. Only five people chose "Other" indicating that academics are the primary users of AI.

Figure8



In figure.8 ChatGPT is the most popular AI app (84%), indicating that consumers strongly prefer it. Gemini and Poe have respective utilization rates of 7%, suggesting that some users look into other AI tools. The majority of users appear to stick to popular options, since only 2% of respondents use other AI applications.

Figure9



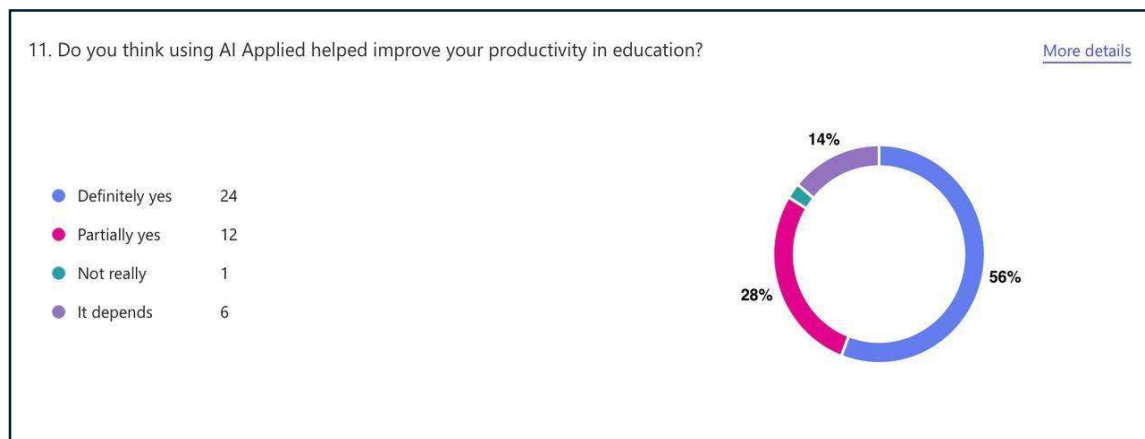
In figure9. AI applications are deemed user-friendly by 89% of participants (Very easy + Easy), suggesting that the majority of users had a seamless experience. 12% are neutral, which means they have no trouble using AI applications. No participant reported difficulty using AI programs, indicating that the user interface and experience are generally simple and easy to use.

Figure10



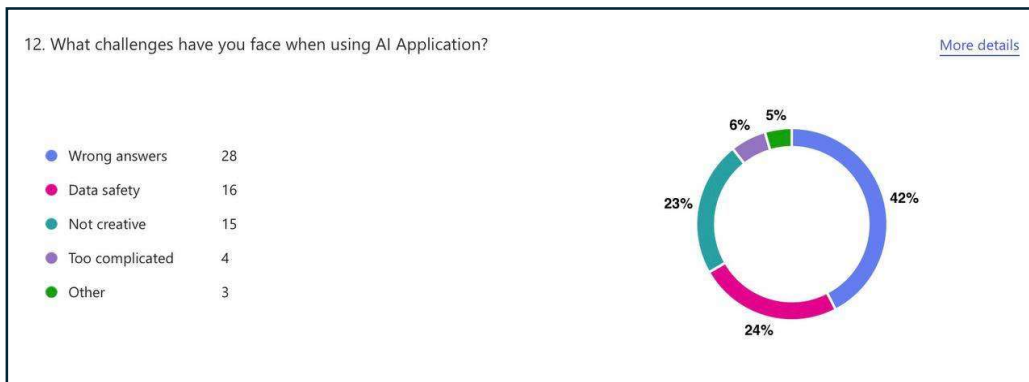
In figure.10 A significant majority (93%) express satisfaction or great satisfaction with the outcomes of AI applications, indicating a high degree of user acceptability. The fact that only 7% of respondents expressed discontent (unsatisfied + Very unsatisfied) suggests that bad experiences are rare. Of those surveyed, 63% are only pleased rather than extremely satisfied, indicating that more work has to be done to raise satisfaction levels.

Figure 11



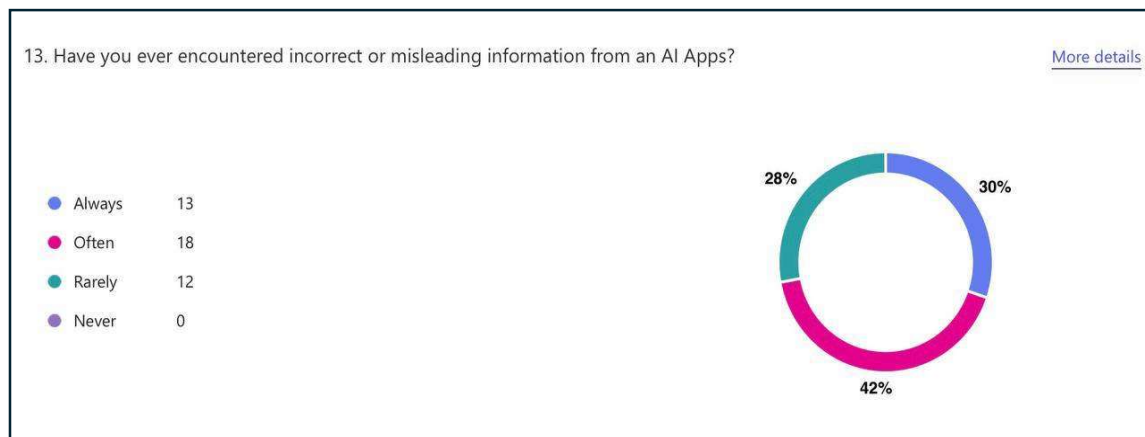
In figure11. Strong support for AI's advantages is demonstrated by the 84% of respondents (Definitely yes + Partially yes) who say it has increased their productivity in the classroom. Only 2% of respondents believe AI has not significantly increased their productivity, indicating that unfavorable opinions are rare. According to 14% of respondents, the influence varies depending on certain aspects, suggesting that the usefulness of AI may differ depending on usage or context.

Figure12



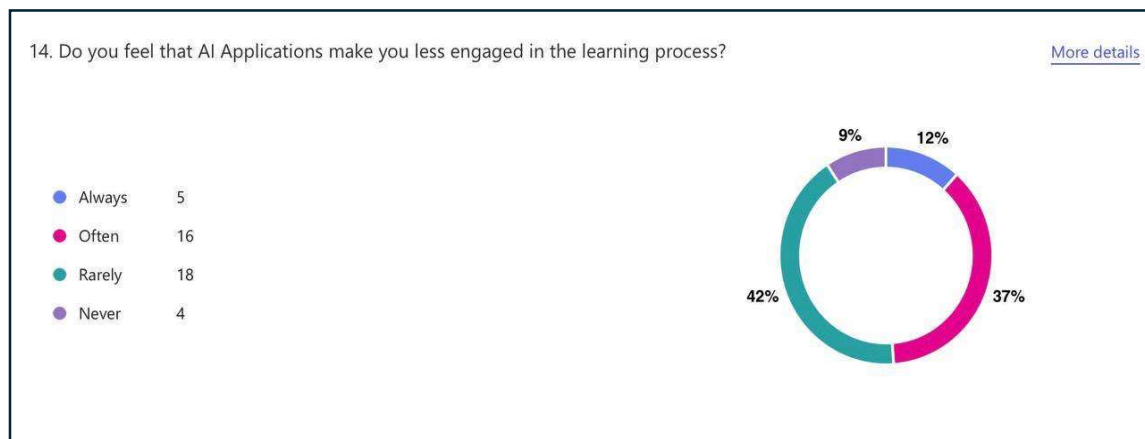
In figure12. Inaccurate responses are the largest issue customers encounter with AI (42%), underscoring concerns about dependability. 24% of consumers are concerned about data security, and 23% believe AI lacks originality. AI applications are largely user-friendly, as evidenced by the fact that only 6% of respondents think they are too complicated. Moreover, 5% brought up other, uncommon problems.

Figure13



In figure13. There is a serious problem with AI reliability, since 72% of participants (Always + Often) reported often receiving inaccurate or deceptive information from AI programs.28% of respondents said they "rarely" came across false information, indicating that although AI mistakes can occur, some users do not constantly experience them. No respondent chose "Never," indicating that all of them have encountered some degree of false information from A applications.

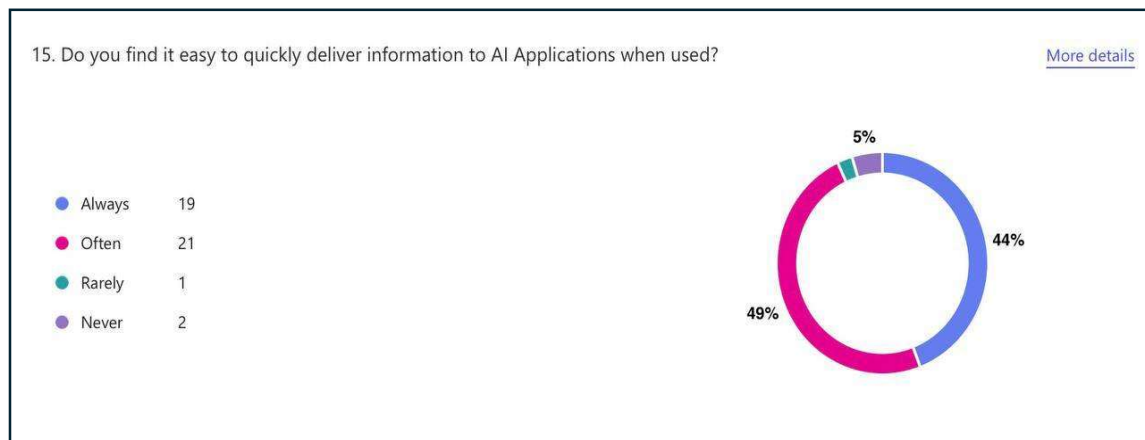
Figure14



In figure14

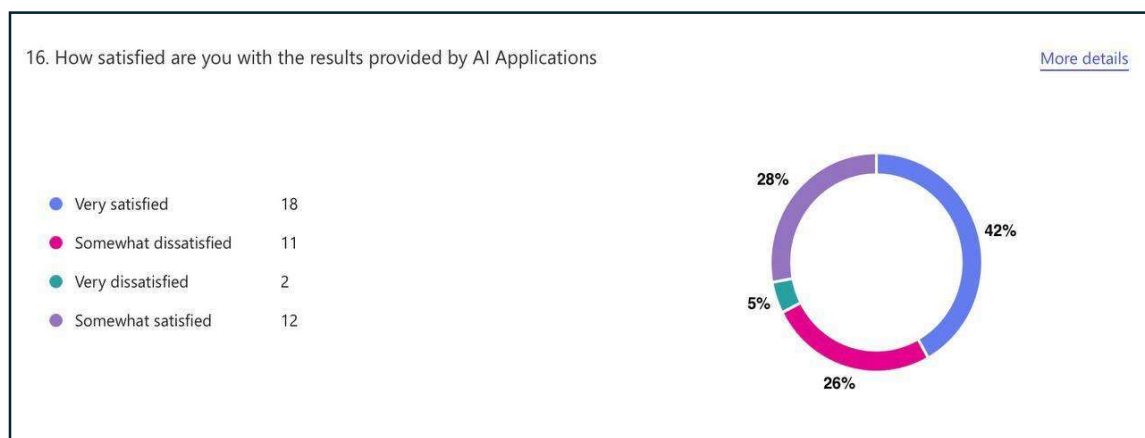
There may be a worry about AI influencing active involvement, as 49% of participants (Always + Often) believe that AI apps lower their engagement in the learning process. 42% of respondents said, "Rarely," indicating that AI has little effect on the engagement of many users. Almost all respondents report feeling disengaged with AI in learning to some extent, with only 9% saying it never impairs their interest.

Figure15



In figure15. Delivering information to AI apps is easy for the vast majority of participants (93%)-44% said "Always," and 49% said "Often." Just 7% of respondents claim they have trouble entering data, with 5% responding "Never" and 2% responding "Rarely." The findings imply that, in terms of data entry and interaction, AI apps are generally easy to use.

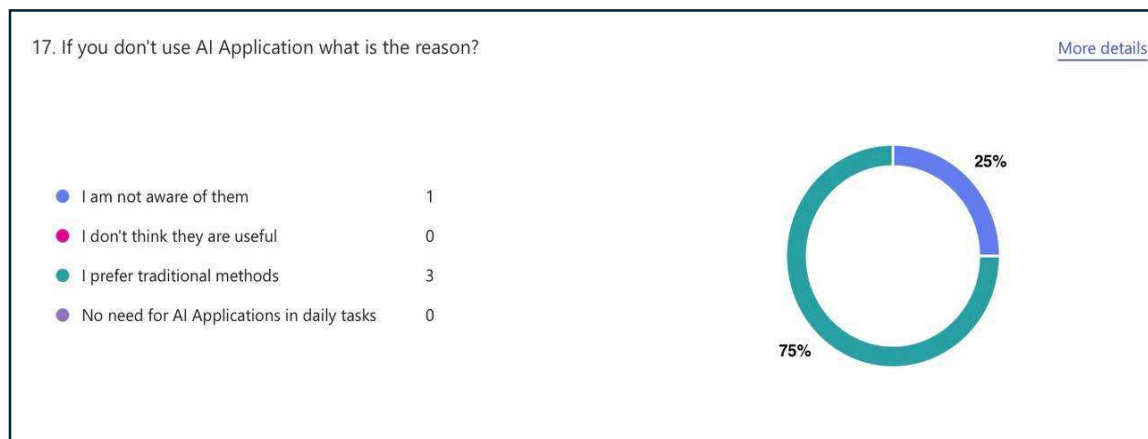
Figure16



In figure16. The vast majority of participants (42%) have a very positive opinion of the outcomes of AI applications and are extremely delighted with them. 28% express some degree of satisfaction, indicating that 70% of

respondents have an overall favorable opinion of AI applications. Of those who find some value, 26% are somewhat dissatisfied, suggesting that they may have run into problems or constraints. Only 5% express extreme unhappiness, indicating that high levels of discontent are uncommon.

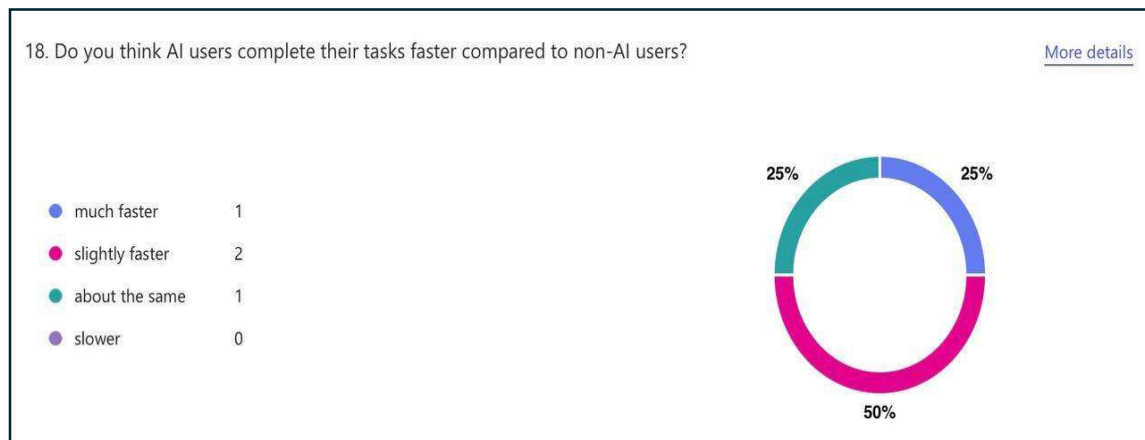
Figure17



In figure17

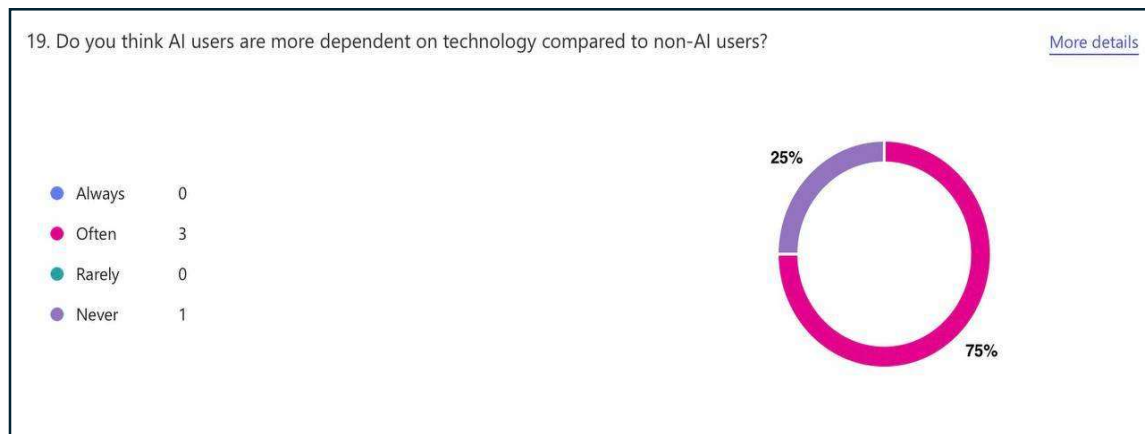
A dependence on manual or recognizable procedures is shown in the 75% of people who prefer using traditional methods over AI apps. A quarter of participants don't know about AI applications, which emphasizes the need for further education and awareness about these technologies. There is a generally neutral or positive view on AI applications' usefulness, since none of the participants think they are pointless or superfluous for everyday work.

Figure18



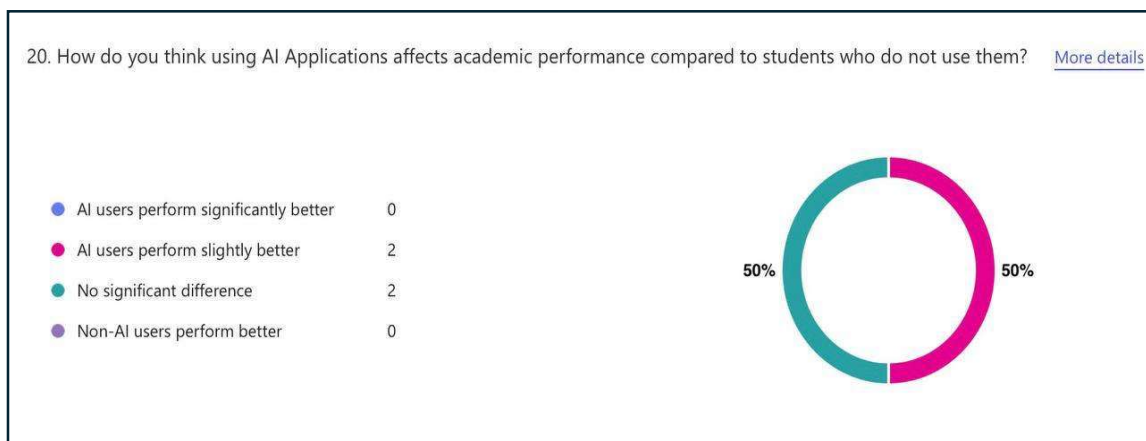
In figure18. There is general agreement that AI improves efficiency, as 75% of participants think that AI users finish their work either much or somewhat faster. 25% believe that the speed of AI and non-AI users is roughly equal, indicating that the use of AI may not have a major impact on particular jobs. The fact that there were no answers for "Slower" suggests that AI is not thought to be a factor that slows down task completion.

Figure19



In figure.19 There is a high association between the use of AI and reliance on technical tools, as 75% of participants think that AI users regularly rely on technology. A smaller percentage of respondents—just 25% believe that AI users do not depend on technology. The lack of answers for "Always" or "Rarely." indicates that people who use AI view their dependence on technology as normal, but neither absolute nor rare.

Figure20



In figure.20 Those who say there is no discernible difference (50%) and those who think AI users have a minor advantage (50%) are equally distributed. The notion that AI users do noticeably better is not well supported. Furthermore, nobody thinks that people who don't use AI perform better, suggesting that AI is not seen as a bad thing for academic achievement.

10. Conclusion

In this study, we focused on understanding how students deal with AI applications in their studies, and what they think about them. Through the answers we have collected, we noticed that most students see that AI helps them a lot in their studies, whether in saving time or facilitating access to information. But at the same time, some of them faced problems such as errors in the answers or concern about the subject of data security.

We have also noticed that in students who feel that AI Applications makes them less interactive with academic content, while others see that it improves their academic level compared to students who do not use it. There was a difference of views, but in general, most students have a positive experience with AI Applications, even if there are some challenges that need solutions.

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